

Is your Body Euhydrated?

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Maintenance of water levels in body is referred as Euhydration. For a normal person the water losses from body generally take place through kidneys in the form of urine, through lungs during respiration, through feces and through skin in form of indispensable water losses as well as visible sweating (McArdle, 1996). These losses are substituted by fluid consumed through food (1000ml), liquid intake (1300ml) and water produced during metabolism (300ml) in our body (Astrand and Rodahl, 1986). The fluid balance is maintained through kidney and very fine tuned hormonal control. If the water/fluid intake is lower than losses, kidney produces less volume of urine and vice versa. In athletes fluid losses through skin and lungs are greater than a normal person and environmental factors like temperature, humidity, wind flow, clothes worn during training can significantly affect the overall fluid losses from the body. Inadequate consumption of fluids during and post training can lead to hypo-hydration. Hypo-hydration can be defined as the negative water balance (water deficit) in the body. The term 'dehydration' is the process of water losses from body (Greenleaf, 1992) however it is often used interchangeably for hypo-

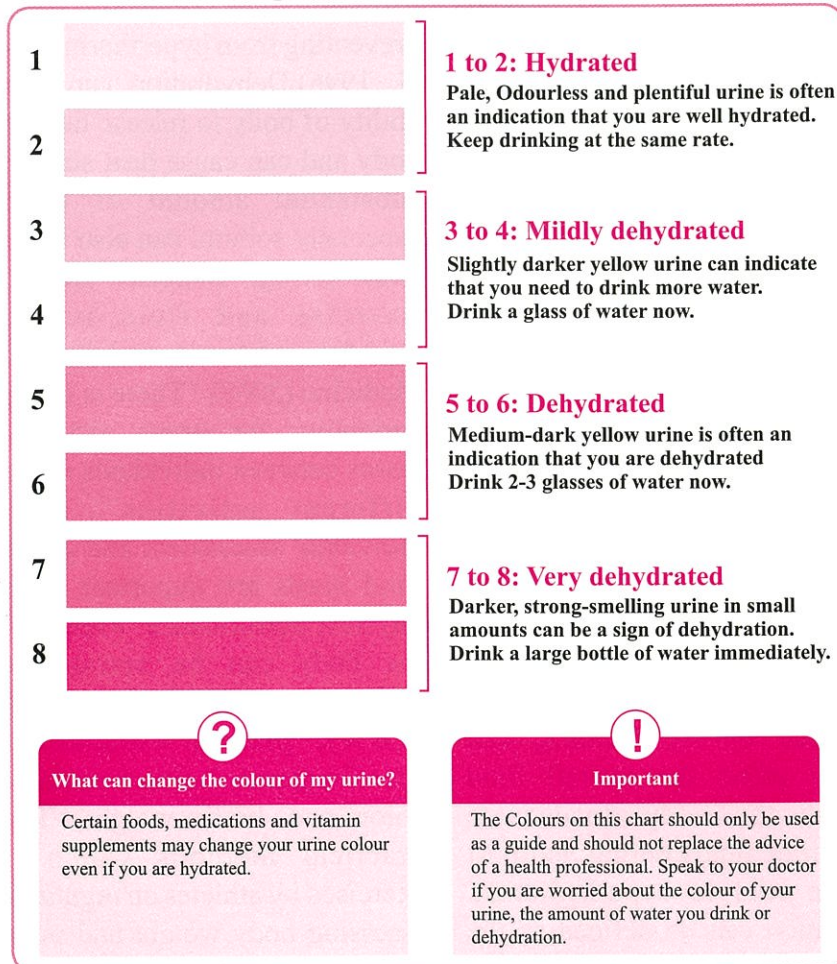
hydration.

During exercise due to metabolic activities heat is produced in the body and sweat helps in removing the excessive heat from our body, thus preventing from hyperthermia (Swaka et al, 1996). Dehydration can impair the ability of body to release this heat from body and can cause heat stress. Further substantial amount of electrolytes especially sodium can also be lost from body during exercise. According to Exercise and fluid replacement guidelines of American college of Sports Medicine (2007) "There is considerable variability for water and electrolyte losses between individuals and between different activities". Therefore individual assessment and guidance for fluid needs are important to maintain physical performance. Fluid balance in the body can be assessed through different methods like total body water (TBW), plasma osmolality, urine specific gravity, urine osmolality, urine color and body weight. The most practical methods which can be exercised by athletes on regular basis are assessing body weight and urine color. Body weight can be assessed pre and post training with minimal dry clothing. The weight loss per gram should be replaced with 1-1.5 ml of water post training to rehydrate the body.

Monitoring urine color over a period of time can provide you a picture of fluctuations in hydration status. Darker the urine color, higher the hypohydration. However consumption of vitamins and mineral supplements

especially B-Complex vitamins can give a false yellow color to the urine. Therefore athletes consuming these supplements should check urine color atleast after 7-8 hours of consuming such supplements.

Figure 1: Urine Color Chart



Source: www.healthdirect.gov.au

Calculating sweat rate can help in forming strategies for individual requirements of fluid during training (International Olympic Committee, 2012). Sweat loss can be calculated using the following formula: Sweat loss

= (Body mass before exercise - Body mass after exercise in kg) + weight of fluids/foods consumed (ml/kg) – Urine losses (ml). Further to calculate sweat rate per hour, it should be divided by the exercise time in minutes and multiply by

60. Dehydration of more than 2% during training or competition poses a detrimental effect on exercise performance (ACSM, 2007). Calculation to derive % dehydration are as follows: $[\text{Weight deficit during training (Kg)} / \text{Starting body mass (kg)}] * 100$. An athlete should drink enough amount of water/fluid to maintain the body weight losses less than 2% during training.

Many times athletes of weight category sports intentionally restrict fluid intake and increase fluid excretion through excessive sweating to achieve weight for a specific weight category. On the other side dehydration can occur due to ignorance and lack of knowledge regarding fluid intake during training and competition in athletes of different sports especially endurance events. Along with hyperthermia dehydration can impair the functioning of cardiovascular system (blood pressure, blood flow, oxygen delivery, metabolite removal), central nervous system (cerebral metabolism, neurotransmitter levels, temperature), peripheral muscles (temperature, metabolic, afferent feedback) and respiratory system (hypocapnia, alkalosis, breathing sensations) (Burke, 2015). Cognitive/mental performance can also be degraded due to dehydration in the body (Hancock and Vasmatazidis, 2003).

Though intake of plain water can suffice for activities lasting for maximum 1 hour, carbohydrate and electrolytes provide added benefits for training/competition of longer

durations. Addition of 6–8% carbohydrate helps in better absorption of water during training as well as provides fuel for energy during training (ACSM, 2007). The 6-8% solution can be prepared by adding 30-40g of glucose/table sugar/honey (6-8 teaspoon) in 500ml of water. Similar to water losses through sweat, sodium losses also vary from individual to individual. High sodium losses in sweat can be checked through “salty sweater” method. Athlete can wear a black t shirt during training and salt stains (white powder) under arms and on the chest post training is indicator of “salty sweater” (IOC, 2012). Athletes training for more than two hour, in hot and humid environment with 'salty sweater' should select sports drinks (10–30 mmol/L sodium) or oral rehydration solutions (range 30–90 mmol/L sodium) to replace the sodium losses during training (Burke, 2015). Drinking plain water while there are excessive sodium losses can lead to exercise associated hyponatremia (serum sodium concentration <130 mmol/L) (ACSM, 2007).

Thus through following simple steps in the training routine, athlete can ensure proper hydration which will ultimately help him/her in achieving desired training goals. Athletes of endurance sports, team sports and weight category sports who train in hot and humid conditions are at higher risk of getting dehydrated. Therefore, individual hydration strategies should be planned for such athletes based on individual tolerance and adaptations.

References

- Astrand PO and Rodahl K. (1986). Textbook of work physiology. Physiological basis of exercise. Third Edition. Singapore: McGraw Hill.
- Burke L (2015). Clinical Sports Nutrition, 5th Edition. McGraw-Hill Australia, 09/2015. VitalBook file.
- Greenleaf JE (1992). Problem: thirst, drinking behaviour, and involuntary dehydration. *Med. Sci. Sports Exerc.* 24, 645–656.
- Hancock PA and Vasmatazidis I (2003). Effects of heat stress on cognitive performance: the current state of knowledge. *Int. J. Hyperthermia* 19:355–372.
- McArdle WD, Katch FI and Katch VL (1996). Exercise Physiology: Energy, Nutrition, and Human Performance. p 54. Philadelphia: Lippincott Williams & Wilkins.
- Nutrition for Athletes: Athletes' medical information. A practical guide to eating for health and performance (2012). Nutrition Working Group of the International Olympic Committee.
- Sawka MN, Wenger CB and Pandolf KB (1996). Thermoregulatory responses to acute exercise- heat stress and heat acclimation. In: *Handbook of Physiology, Section 4: Environmental Physiology*, Blatteis CM and Fregly MJ. p 157–186. New York: Oxford University Press for the American Physiological Society.
- Urine Color Chart. <https://www.healthdirect.gov.au/urine-colour-chart>. Retrieved on 2nd January, 2020.